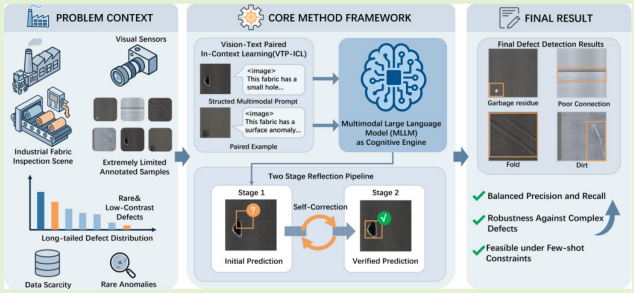
Few-Shot Fabric Defect Detection via Vision-Text Paired In-Context Learning and Two-Stage Reflection

Zhizhi Peng, Zhenfang Liu, Tongzhen Xing, Qing Shen, Zhen Wang, Changhong He, Jungang Lou, *Senior Member, IEEE*, and Zhen Bi

***Abstract*— Industrial visual sensing systems play a critical role in automated quality inspection. However, deploying data-driven perception models on industrial visual sensors faces significant bottlenecks due to the extreme scarcity of annotated anomaly samples and the pronounced long-tailed distribution of defect types. These factors render traditional supervised models prone to overfitting and catastrophic failure when sensing rare anomalies. To address these limitations in the sensor data processing pipeline, this article proposes a novel few-shot perception framework leveraging multimodal large language models (MLLMs) as a cognitive backend, without task-specific fine-tuning of the sensing model. Our approach introduces two key innovations: 1) a vision-text paired in-context learning (VTP-ICL) mechanism that constructs structured multimodal prompts to activate pretrained knowledge for precise defect localization and 2) a two-stage reflection pipeline that decouples initial prediction from verification, enabling the model to self-correct outputs based on explicit criteria. Extensive experiments on a real-world industrial fabric dataset demonstrate that our method achieves state-of-the-art performance under strict few-shot conditions (two shots per class). Notably, our approach maintains an exceptional balance between precision and recall, significantly outperforming traditional detectors, particularly on hard-to-sense, low-contrast defects. These results validate the potential of MLLMs as robust and generalizable engines for next-generation industrial sensing applications.**



***Index Terms*— Few-shot learning, in-context learning (ICL), industrial sensing, multimodal large language models (MLLMs), sensor data processing.**

I. Introduction

INDUSTRIAL visual inspection is a pivotal application domain of modern sensing technology, ensuring product quality and manufacturing efficiency. Among various sensing modalities, vision-based sensors are widely deployed to capture the intricate details of production lines. However, fabric defect inspection presents a particularly challenging scenario for these sensors due to the intricate periodic textures of textiles, where anomalies often exhibit low contrast and high similarity to the background [1], [2], [3]. Moreover, the diversity of fabric materials and weaving patterns exacerbates the difficulty of building generalizable sensing models [4], [5]. From a sensing perspective, these characteristics cause traditional computer vision approaches to exhibit suboptimal performance in real-world textile quality control.

A fundamental bottleneck in deploying data-driven solutions for fabric inspection lies in the limitations of sensing data acquisition. Industrial production lines are optimized for "zero-defect" output, making genuine anomalies rare and costly to collect [6], [7]. Precise annotation of subtle fabric flaws demands significant domain expertise and manual effort [8], resulting in datasets that are not only small in scale but also severely limited in class coverage. This scarcity confines the problem to a few-shot regime, where conventional supervised models struggle to generalize due to overfitting [9], [10]. Compounding this data scarcity is a pronounced difficulty imbalance across defect categories. Real-world fabric defects

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follow a long-tailed distribution: common and visually salient defects may appear with relatively higher frequency, while critical yet subtle anomalies occur extremely rarely and are easily overlooked [11], [12]. Models trained on such imbalanced data tend to perform well on easy, majority classes but fail catastrophically on hard, minority ones—a critical flaw in industrial settings where missing a rare defect can have severe consequences. Consequently, there is an urgent need for intelligent sensing systems that can achieve high-fidelity perception with minimal calibration data.

To address these intertwined challenges of extreme data scarcity and intraclass difficulty imbalance, we propose a novel few-shot fabric defect perception framework based on a multimodal large language model (MLLM). Our approach treats the MLLM as a cognitive processing unit for the visual sensor, leveraging rich priors and reasoning capabilities without requiring task-specific fine-tuning. Specifically, 1) we introduce a vision-text paired in-context learning (VTP-ICL) mechanism, designed to address the few-shot problem by constructing structured multimodal prompts to effectively activate the model's pretrained knowledge and 2) we design a two-stage reflection pipeline that decouples initial prediction from verification, enabling the model to self-correct its outputs based on explicit criteria. In summary, our contributions are threefold.

1) We present one of the first MLLM-based frameworks tailored for few-shot fabric defect sensing, explicitly tackling the dual challenges of scarce annotations and imbalanced difficulty.
2) We propose VTP-ICL for effective context utilization and a reflection mechanism (RM) with constrained reasoning to enhance robustness on hard examples.
3) Extensive experiments demonstrate that our method achieves state-of-the-art performance with exceptional balance between precision and recall, significantly outperforming traditional visual models under strict few-shot conditions.

II. Related Work

A. Industrial Anomaly Detection Model

In the field of industrial defect detection, visual models based on convolutional neural networks (CNNs), such as YOLO and Faster R-CNN, have achieved remarkable results in general object detection. However, they inherently rely on large-scale, high-quality, and class-balanced annotated data, limiting their performance in scenarios demanding "massive data" for supervised learning [13]. Traditional methods exhibit weak transferability when annotated data is scarce, with generalization ability degrading sharply in few-shot scenarios [14]. This issue is particularly prominent in industrial settings where defect samples are naturally scarce, costly to obtain, and lead to extreme data imbalance [15], [16]. Consequently, traditional supervised models are prone to overfitting; when anomaly samples are extremely limited, these models tend to memorize training data rather than learn generalized features, impairing test set robustness [17]. For example, in solar cell quality inspection, traditional models perform poorly in few-shot tasks despite data augmentation, failing to effectively improve generalization [18], [19]. Furthermore, industrial defect types are open and dynamically evolving, whereas traditional models assume consistent training and test distributions, leading to significant degradation against unseen defects [20]. Combined with issues like unstable data quality and noise interference [21], the scarcity of anomaly samples and high annotation costs restrict the applicability of the supervised learning paradigm [22]. Thus, while CNN detectors excel under ideal conditions, their dependence on large datasets, overfitting tendencies, and weak adaptability to new defects have driven researchers toward emerging paradigms like few-shot learning [23], [24].

B. Application of MLLMs in Industrial Scenarios

In recent years, MLLMs have broken through the limitations of traditional perception tasks in industrial scenarios. Acting as a "cognitive engine," their advantages in semantic understanding, instruction following, in-context learning (ICL), and logical reasoning provide a new paradigm for solving core challenges such as scarce data, complex rules, and high deployment costs. Regarding semantic understanding and instruction following, MLLMs can convert unstructured natural language instructions into executable industrial logic: SayComply maps operation manuals into robot compliant action sequences via retrieval-augmented models [25]; customizing visual-language foundation models allows engineers to dynamically define anomaly detection standards in natural language, enhancing human–machine collaboration [26]; and TL-CLIP constructs a multimodal model aligned with professional vocabulary for power inspection, realizing reliable associations between terms and visual signals [27]. To address the scarcity of annotated samples, MLLMs achieve efficient few-shot/zero-shot transfer through ICL: IADGPT proposes a unified framework capable of identifying, localizing, and attributing anomalies on new production lines with only 1–5 examples [28]; PromptAD generates "ideal state" text prompts from normal samples to guide visual models [29]; while FADE and anomaly-OV verify that general visual-language models can achieve anomaly perception through semantic priors without accessing target domain data, effectively solving the industrial cold start problem [30], [31]. More importantly, the reasoning ability of MLLMs enables the explanation of anomaly causes: LogicAD combines MLLM text features with formal logical rules to discover "logical-level" anomalies through automatic theorem proving [32]; Reason-IAD introduces an expert knowledge graph to drive causal reasoning for root cause analysis [33]; and AnomalyR1 and EMIT optimize reasoning chains via reinforcement learning to improve judgment robustness for subtle and occluded defects [34], [35]. This represents a cognitive advantage difficult for traditional methods to achieve. Ultimately, the industrial value of MLLMs lies in their comprehensive intelligence—integrating semantic understanding, few-shot adaptation, and causal reasoning, which lays the foundation for building highly generalizable,

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interpretable, and easy-to-deploy systems for few-shot defect detection.

III. Methodology

A. Task Definition

The fabric defect detection task can be formulated as a multimodal visual grounding task. Given an input image $I \in \mathbb{R}^{H\times W\times 3}$ and a text instruction T, the goal of the model is to output a set of 2-D bounding boxes for defect targets

$$B = \{b_i = (x_{i1}, y_{i1}, x_{i2}, y_{i2}) \mid i = 1, 2, \ldots, N\} \tag{1}$$

and the corresponding set of category labels

$$L = \{l_i \mid i = 1, 2, \ldots, N\} \tag{2}$$

where b_i represents the bounding box of the ith defect target output by the model, i is the serial number of the defect target, with a value range of $1, 2, \ldots, N$; N represents the total number of defect targets detected in a single image; x_{i1}, y_{i1} are the top-left pixel coordinates of the ith defect bounding box, and x_{i2}, y_{i2} are the bottom-right pixel coordinates of the corresponding bounding box. All coordinates are defined based on the image size and take nonnegative integer values. l_i represents the defect category label corresponding to the ith defect bounding box b_i, with the label value taken from a predefined set of defect categories.

Therefore, the entire detection task can be expressed as the following mapping function:

$$f_\theta : (I, T) \mapsto \{(b_i, l_i)\}_{i=1}^{N} \tag{3}$$

where f_θ denotes an MLLM with parameters θ.

This setting requires the model to achieve accurate localization and classification of subtle, low-contrast, or morphologically variable defects in complex texture backgrounds. This not only tests the model's cross-modal understanding ability but also places requirements on its fine-grained perception, spatial reasoning, and semantic consistency judgment.

B. Visual Demonstration

In this section, we propose the VTP-ICL mechanism. Central to this approach is ICL, an emergent ability of large models that allows them to rapidly learn and generalize novel concepts solely through conversational examples in the prompt, without any parameter updates. Unlike traditional fine-tuning, ICL allows the model to infer task-specific patterns and execute complex reasoning solely based on the provided context, without modifying its internal parameters. It is crucial to clarify that when we refer to "activating pretrained knowledge," we do not imply that the model has explicitly seen specific fabric defects during pretraining. Instead, the pretraining phase establishes a robust cognitive foundation that enables this ICL capability. By leveraging the conversational paradigm to provide structured visual–textual examples, we exploit this underlying learning ability, allowing the model to rapidly generalize to unseen defect patterns without parameter updates.

In the few-shot fabric defect detection task, text-only prompts fail to fully capture the mapping between complex visual patterns and spatial semantics. To address this, VTP-ICL activates multimodal priors by embedding example images paired with structured descriptions—specifically bounding box coordinates and semantic labels—into the prompt context. These visual annotations serve as explicit spatial cues bridging low-level features and high-level semantics, enabling the model to establish an interpretable correspondence between visual defects and their descriptors.

Specifically, for each defect example image I_k before inputting it into the model, we perform visual annotation on the defect region: the defective area is highlighted with a distinct bounding box overlay (in a high-contrast color space) and a category label tag at the top-left corner of the box, which is consistent with the spatial position and semantic information of the subsequent text-based [coordinate, label] pairs. This visual annotation strategy follows the principle of prompt consistency in multimodal learning, ensuring that the visual signal and textual signal received by the model are semantically and spatially aligned, and avoiding the modality misalignment problem that may arise when the model maps unannotated visual features to discrete text coordinates. On this basis, we construct a few-shot example context

$$C = \left\{\left(I_k^{(1)}, y_k^{(1)}\right), \left(I_k^{(2)}, y_k^{(2)}\right)\right\}_{k=1}^{K} \tag{4}$$

containing K types of defects, where each sample consists of an image $I_k^{(i)}$ and the corresponding structured annotation

$$y_k^{(i)} = \left\{\text{bbox} : \left[x_1, y_1, x_2, y_2\right], \text{label} : c_k\right\} \tag{5}$$

where $c_k \in L$ is a predefined set of defect categories. For each defect example image I_k of a category, we first input it alone into an MLLM with frozen parameters to guide the model to generate a natural language description d_k. Subsequently, we embed the original example image I_k paired with its corresponding model-self-generated descriptiond_k (as well as precise bounding box coordinates and category labels) into the final prompt context

$$T = \bigoplus_{k=1}^{K}\left(I_k^{(1)} \oplus y_k^{(1)} \oplus I_k^{(2)} \oplus y_k^{(2)} \oplus d_k\right) \oplus I \oplus [\text{QUERY}] \tag{6}$$

where $\oplus$ denotes a modality-aware sequence concatenation operation, which is implemented by the MLLM through a multimodal connector during inference; $[QUERY]$ represents the query instruction for the task to be detected, which is a standardized reasoning prompt for the model. It is used to clarify the core task requirements for the current fabric defect detection (such as locating defects, labeling categories, and outputting structured coordinates), guiding the model to focus on the image to be tested I for detection reasoning. The core of this prompt structure lies in a triple in-context induction mechanism.

1) *Visual Annotation Induction:* The highlighted bounding box and label on the example image serve as visual prompts, preactivating the model's attention to defect regions. This focuses perception on specific defects

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rather than the complex background, reducing cognitive load and improving visual–textual alignment efficiency.

2) *Spatial Reference Induction:* Embedding precise 2-D coordinates provides an intuitive spatial reference. This enables the model to dynamically align visually annotated regions with coordinate descriptions, ensuring geometrically consistent predictions.
3) *Semantic Anchoring:* The category label c_k and description d_k, generated from the annotated image, form a strong semantic anchor. As d_k derives directly from the model's perception of I_k, it aligns naturally with visual patterns. This "image-to-text" approach captures discriminative features accurately, locating defect prototypes in semantic space and enhancing robustness against low-contrast or deformed defects.

In summary, without introducing new learning objectives, this structured context construction efficiently schedules the model's existing multimodal priors. This enables effective fabric defect detection under zero-shot or few-shot conditions without fine-tuning. The key insight is that structuring context as image–text pairs aligns with the human expert paradigm of "image-to-text" reasoning. Embedding this into the inference context facilitates precise localization of subtle defects in unseen samples, strictly adhering to the constraint of zero gradient updates.

C. Instruction Constraints for Structured Output

Multimodal large models generally tend to generate natural language explanations in open-domain tasks, but industrial quality inspection systems require detection results to be highly deterministic and machine-parsable. To address this, we have designed a set of strict instruction constraint mechanisms aimed at guiding the model to abandon free-text output and instead generate structured results that comply with the specifications of downstream automated processing. The core of this mechanism lies in Format-Aligned Prompt Engineering, which imposes explicit format restrictions on the model's output behavior. During inference, the model is explicitly instructed to output in a specified JSON format after identifying defective regions in the image, with the specific form defined in (5). The instructions specifically emphasize that no additional text, explanatory sentences, or unstructured content shall be included, thereby ensuring that the output results can be directly parsed by programs and avoiding system integration failures caused by inconsistent formats.

To ensure the accuracy and physical significance of coordinate prediction, we explicitly inform the model of the specific dimensions of the current input image in the prompt (e.g., "the image has a width of W pixels and a height of H pixels"). This information enables the model to perform spatial reasoning in a real pixel coordinate system, rather than relying on vague relative position estimation. Meanwhile, the instructions further require that all bounding box coordinates must be accurately given in integer form, prohibiting the use of vague expressions such as "about" or "approximately" or rounding operations. This strict specification on numerical accuracy effectively improves the reliability of positioning results, allowing them to be directly used for subsequent industrial operations such as cropping, marking, or repair.

This instruction constraint mechanism deeply integrates image scale priors with format specifications, constructing a reliable channel from perception to parsable output, and providing practical support for the engineering implementation of the entire detection process.

D. Reflection Mechanism

MLLMs possess powerful semantic reasoning capabilities, yet existing related research has demonstrated that explicit textual reasoning [e.g., chain-of-thought (CoT)] is detrimental to fine-grained visual tasks requiring precise coordinate output, such as target detection. This limitation stems from the inherent mismatch between CoT's reliance on descriptive textual reasoning and the rigorous numerical nature of spatial localization: as documented in recent studies [36], introducing semantic reasoning pathways tends to inject noisy estimative language into the output, which impairs the model's fine-grained spatial perception fidelity and leads to systematic deviations in bounding box coordinates.

Against this background, a core challenge emerges: how to effectively harness the inherent reasoning potential of MLLMs without compromising the precision of numerical detection outputs? To resolve this contradiction, we propose a two-stage reflection pipeline, fundamentally designed as a multiturn conversational process that strategically decouples coordinate generation from semantic verification. This "intuitive output first, result reflection later" framework addresses the inherent conflict between CoT reasoning and precise spatial localization through two distinct operational stages. By avoiding the drawbacks of unconstrained explicit reasoning during initial prediction while activating deferred self-correction, it strikes a balance between perceptual accuracy and reasoning-based verification, thereby optimizing detection robustness for subtle, low-contrast fabric defects prone to misjudgment. Specifically, the pipeline operates as follows.

Stage 1 (Direct Perception via Nonreasoning Mode): In the first dialog turn, the model operates in nonthinking mode to generate initial detections. Unlike logical tasks, bounding box coordinates are nonderivative atomic outputs relying on direct visual-language mapping. Crucially, in-context examples provide precise position annotations [see (6)] as high-fidelity demonstrations. Disabling thinking mode bypasses intermediate textual reasoning, preventing estimative tokens (e.g., "approximately") from contaminating the token sampling distribution and biasing outputs toward coarse-grained values. This ensures pixel-level precision anchored by the in-context demonstrations.

Stage 2 (Anchored Self-Correction via Deferred Reasoning): In the second turn, we activate thinking mode for self-correction. The full conversational history—base context, demonstrations, and Stage 1 results—will be fed back with an explicit instruction to verify previous detections. This reflection is anchored by Stage 1's high-fidelity coordinates; since the solution space is already constrained by precise tokens, cognitive resources shift from spatial estimation to logical validation (e.g., category consistency and spatial plausibility).

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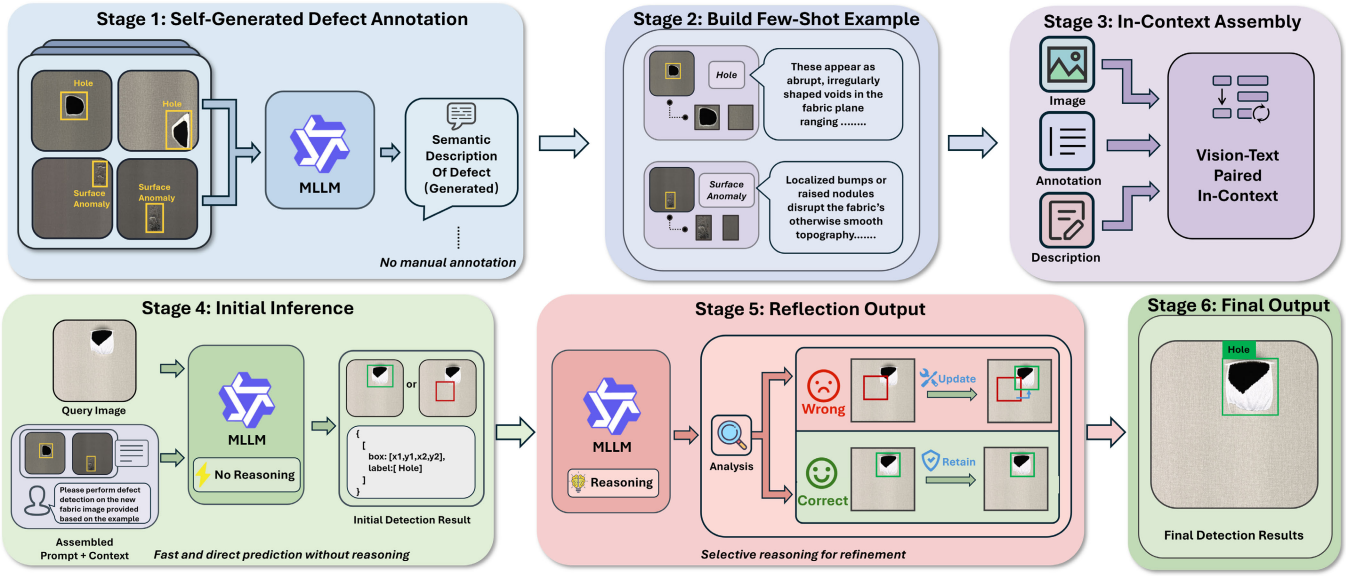


Fig. 1. Overview of the proposed VTP-ICL framework with two-stage reasoning and RM. The pipeline comprises six stages, all built upon a single MLLM. The model first performs self-generated annotation based on the provided defect sample images to produce semantic defect descriptions, which are then used to construct few-shot examples and assemble vision-text in-context prompts. It subsequently generates initial predictions without reasoning, followed by a reflection stage with reasoning enabled to analyze and refine initial predictions. The final stage outputs improved detection results. This unified framework mitigates the degradation of output precision caused by token space contamination from premature reasoning by deferring the use of reasoning to a dedicated reflection stage, thereby enabling more effective and targeted utilization of the model's reasoning capability for prediction refinement.

This deferred reasoning enables robust self-correction without regressing to coarse estimation, leveraging CoT for verification while preserving localization accuracy.

Formally, let the output of the first stage be $Y^{(0)} = \{(b_i^{(0)}, l_i^{(0)})\}_{i=1}^{M}$, and the reflection stage performs the following mapping:

$$Y^{(1)} = \text{Reflect}\left(I, Y^{(0)}; C\right) \tag{7}$$

where Reflect denotes the response function of the multimodal model with frozen parameters under a specific reflection prompt, and C is the aforementioned image–text example context.

The key advantage of this design is that it not only avoids the inherent defects of CoT in initial numerical generation but also effectively utilizes the model's thinking ability. Importantly, the effectiveness of the reflection step hinges critically on the fidelity of the initial prediction: if the first-stage output deviates substantially from the ground-truth region, the reflection process may lack sufficient grounding to recover an accurate result. We will conduct a systematic investigation into the root causes of performance degradation induced by explicit textual reasoning, as well as the sensitivity of the model's correction capability to different reflection prompt templates, in the ablation studies presented in Section IV.

E. Complete Process

In summary, the complete workflow of the proposed framework is illustrated in both Fig. 1 and Algorithm 1. The framework unifies vision-text paired context construction with a two-stage RM, detailing the sequential transformation from few-shot prompt assembly to the final refined detection results.

Algorithm 1 Few-Shot Fabric Defect Detection With VTP-ICL and Reflection

Input: Test image I, Few-shot set $\mathcal{C}_{shot} = \{(I_k, y_k)\}_{k=1}^{K}$, Image dims (H, W), MLLM f_θ
Output: Final detection results $Y_{final} = \{(b_i, l_i)\}_{i=1}^{N}$
$T \leftarrow \emptyset$
for $(I_k, y_k) \in \mathcal{C}_{shot}$ **do**
  $d_k \leftarrow f_\theta(I_k, \text{“Describe defect”})$
  $E_k \leftarrow \text{Concat}(I_k, y_k, d_k)$
  $T \leftarrow T \oplus E_k$
end
$Y^{(0)} \leftarrow f_\theta(T)$ // Non-reasoning mode
$T_{ref} \leftarrow T \oplus Y^{(0)} \oplus$ “Review and correct”
$Y^{(1)} \leftarrow f_\theta(T_{ref})$ // Reasoning mode
Parse $Y^{(1)} \rightarrow$ extract boxes/labels
Filter invalid outputs (out-of-bounds/non-integer)
return $Y_{final} \leftarrow Y^{(1)}$

IV. Experiment

A. Evaluation Metrics

To comprehensively evaluate the performance of the proposed method in the few-shot fabric defect detection task, we adopt a multidimensional evaluation system. In addition to standard object detection metrics, two custom metrics are introduced to capture the model's extreme behavioral tendencies under low-sample conditions, and adaptations are made for the output characteristics of MLLMs.

1) Basic Detection Metrics: We first use the commonly used precision (P), recall (R), and $F1$-score in the field of object detection as basic evaluation metrics, and determine the

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correctness of prediction results based on a greedy matching strategy under a fixed IoU threshold.

Specifically, let the prediction set output by the model for a single image be $B = \{(b_i, c_i)\}_{i=1}^{M}$, where b_i is the predicted bounding box and c_i is its category; the ground-truth annotation set is $B^{gt} = \{(b_j^{gt}, c_j^{gt})\}_{j=1}^{N}$. We first calculate the IoU between all prediction-ground truth pairs

$$\mathrm{IoU}\left(b_i, b_j^{gt}\right) = \frac{\left|b_i \cap b_j^{gt}\right|}{\left|b_i \cup b_j^{gt}\right|}. \tag{8}$$

Subsequently, class-aware greedy matching is performed: for each ground-truth target (b_j^{gt}, c_j^{gt}), the prediction box with the maximum IoU that meets the IoU threshold is selected from the model's output predictions as the match. Successful matches are recorded as true positives (TPs); unmatched predictions are regarded as false positives (FPs); ground-truth targets not covered by any prediction are counted as false negatives (FNs).

Since the MLLM output is a sequence of natural language tokens (e.g., "bbox: [210, 305, 280, 360], class: surface damage"), we convert it into a structured format through regular expressions and a coordinate parser. If the output format is incorrect or the coordinates are out of bounds, the prediction is considered invalid (counted as FP).

On this basis, we calculate precision (P), recall (R), and $F1$-score

$$P = \frac{TP}{TP + FP}. \tag{9}$$

Precision refers to the proportion of truly positive targets among all results predicted by the model as "targets," focusing on the reliability of the prediction results

$$R = \frac{TP}{TP + FN}. \tag{10}$$

Recall refers to the proportion of all truly existing targets that are successfully detected by the model, focusing on the ability to cover real targets

$$F1 = 2 \cdot \frac{P \cdot R}{P + R}. \tag{11}$$

The $F1$-score is the harmonic mean of precision and recall, measuring the balance between the model's reliability and coverage.

2) Extreme Behavior Suppression Metrics: Under few-shot settings, the model may adopt extreme detection strategies—for example, only responding to highly typical samples (resulting in high P and low R), or over-detecting to generate a large number of false detections (resulting in low P and high R). Although such behaviors may perform well in a single metric, they seriously damage the reliability of actual deployment.

To this end, we propose the balance index (BI) to quantify the balance of model output

$$\mathrm{BI} = \frac{\min(P, R)}{\max(P, R)}. \tag{12}$$

Its value range is $[0, 1]$, and a value closer to 1 indicates that P and R are more balanced, and the model is less likely to fall into extreme strategies.

Furthermore, we define the balanced performance score (BPS) as

$$\mathrm{BPS} = \mathrm{BI} \times F1. \tag{13}$$

In traditional $F1$-scores, if there is an extreme imbalance between precision and recall, the $F1$-score may be inflated. Therefore, we introduced the BPS metric; this metric jointly measures detection performance and strategy stability. A higher BPS means that the model not only has good overall accuracy (high $F1$) but also maintains a reasonable precision–recall tradeoff (high BI), making it more suitable for scenarios with strict robustness requirements such as industrial quality inspection.

B. Dataset

This study adopts a private fabric defect dataset provided by a cooperative factory. The dataset contains nine types of defects; all images have a unified resolution of 1024×1024, and each image contains only one defect instance, ensuring that the task focuses on single-target localization and classification. To simulate the industrial conditions of "extremely few annotations" in real scenarios, we divide the data as follows.

1) *Training Set:* Only two example images are provided for each category, totaling 18 images, which are used to construct context examples or train traditional models.
2) *Validation Set:* Ten images per category, totaling 90 images, are used for comparative validation during the training process of traditional models.
3) *Test Set:* Most categories contain 40 images, but due to the uneven distribution of actual production line data, two types of defects have only 16 and 17 images, respectively, due to scarce samples, totaling 313 test images.

C. Experimental Setup

1) Traditional Visual Models: We select a variety of classic object detection models as baselines, including YOLOv8 [37], YOLO11 [38], RT-DETR [39], FasterR-CNN [40], and GroundingDINO [41]. All models are trained based on pre-trained weights using open-source code. Crucially, all data augmentation strategies (such as random cropping, color jitter, and rotation) are explicitly turned off to avoid introducing additional supervision signals, thereby ensuring that the evaluation results truly reflect the learning ability of "relying only on two samples."

2) MLLM (Ours): Our method is mainly based on the Qwen3-VL [42] MLLM, and inference is performed through API calls. The prompt template adopts the image–text combined context structure described in Section III-B, providing two support samples (including images, bounding box coordinates, and natural language descriptions) for each category, and adopting the RM described in Section III-C. The API call parameters are shown in the following Table I.

3) Evaluation Protocol and Fairness Statement: It should be noted that, because our approach employs a frozen MLLM, its output consists of discrete token sequences without inherent confidence scores. Consequently, the traditional mAP

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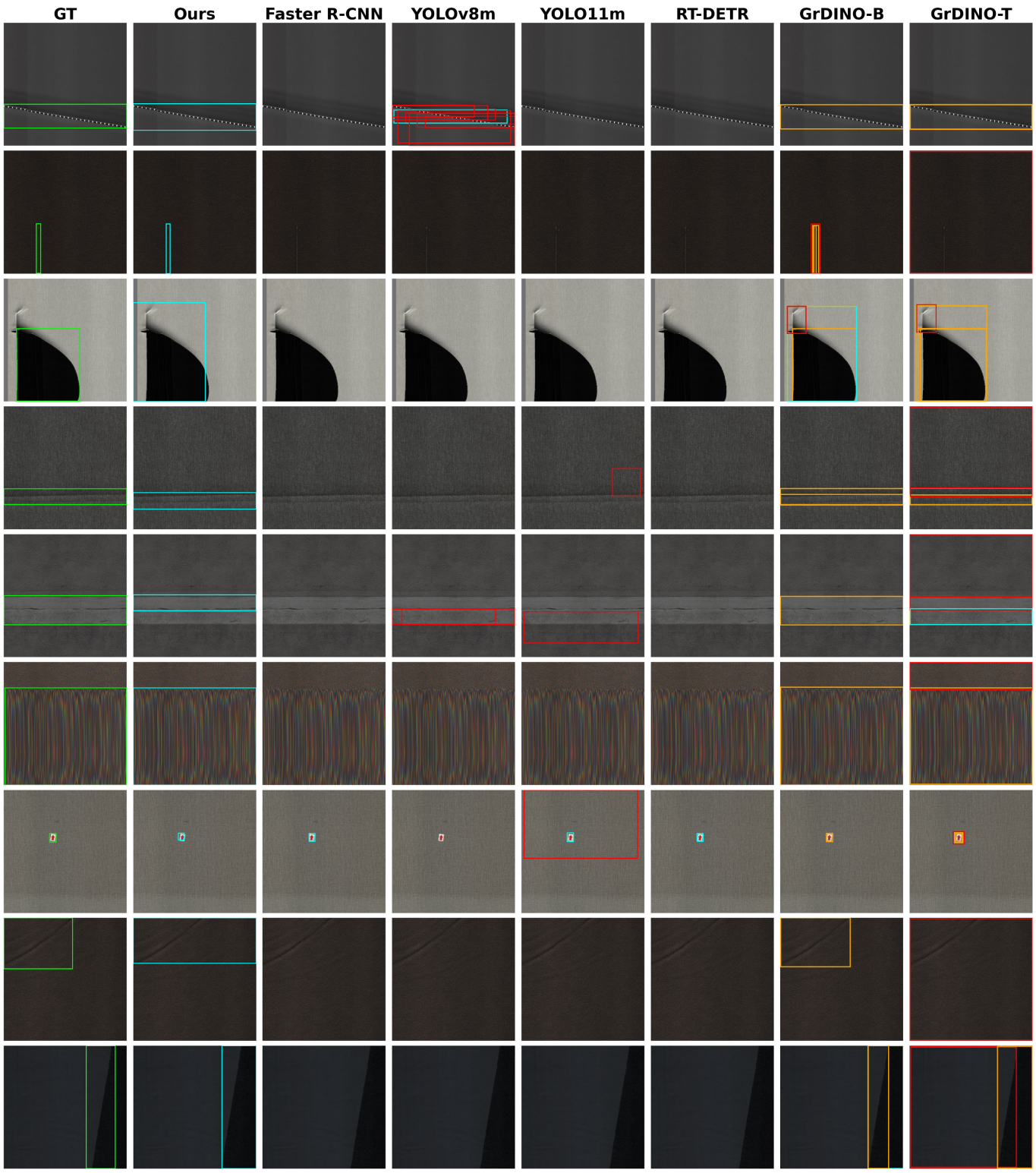


Fig. 2. Comparative visualization of detection results for all nine defect types on representative fabric samples across different models. Color legend: green boxes = ground truth; blue boxes = correct detection (TP); red boxes = incorrect detection (wrong location/duplicate); orange boxes = correct location but wrong label. Most baseline methods suffer from missed detections or false predictions, including duplicates, mislocalizations, or misclassifications. In contrast, our proposed method achieves more consistent alignment with GT annotations, minimizing both FNs and erroneous classifications, and demonstrates superior robustness across diverse defect patterns and textures.

metric, which relies on confidence threshold scanning and AP integration, cannot be directly applied. While confidence could be estimated indirectly via heuristics (e.g., generation probability), such methods introduce additional assumptions that may distort the model's actual behavior. To maintain evaluation consistency with the model's output format, we therefore employ the fixed IoU threshold and greedy matching strategy outlined earlier, foregoing the use of mAP. This leads

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TABLE I
HYPERPARAMETERS FOR MLLM INFERENCE

	temperature	top_p	max_tokens	thinking_budget
Reasoning	0.8	0.95	2048	2048
Non-Reasoning	0.7	0.8	2048	—

[a]Configuration details for the two-stage pipeline: Non-Reasoning stage (Stage 1) focuses on precise coordinate generation, while Reasoning stage (Stage 2/Reflection) utilizes higher temperature for semantic verification.

to an unavoidable asymmetry in evaluating traditional visual models (which output confidence scores) versus MLLMs (which do not). To fairly compare the "limit performance under few-shot conditions," we traverse multiple confidence thresholds for traditional models and report their highest $F1$-scores. Although this introduces a "posthoc optimization" bias, it reveals each model's performance upper bound under experimental constraints. We acknowledge the methodological limitations of this approach but consider it a necessary compromise to balance comparability and practicality given the current inability of MLLMs to produce confidence outputs.

D. Main Results

We conducted experiments based on the above settings. All metrics are calculated separately based on three different IoU thresholds (0.5, 0.75, and 0.9), using a class-aware greedy matching strategy. For each threshold, we independently perform matching and counting, and finally obtain three sets of complete evaluation results for all samples (see Table II) and hard samples (see Table III). The hard samples are screened based on the following criteria: taking the output results of YOLOv8m-pt (the model with the most balanced P and R among traditional closed-set detectors) as the benchmark, we calculate the $F1$-scores of the model for the nine defect types in the test set, and select the four types with the lowest $F1$-scores to form a difficult sample dataset. This screening strategy is consistent with the definition of hard samples in this article—subtle, low-contrast, and morphologically variable defects that are difficult to detect—and effectively simulates the industrial scenario where a small number of defect types account for the majority of detection errors. However, visualized the detection results of all models for a set of nine different defect types in Fig. 2.

1) Closed-Set Detection Models: Moderate Precision With Severe Missed Detections: Under few-shot conditions, closed-set models show a typical trend of "high precision and low recall." Although Faster R-CNN achieves the highest precision at IoU > 0.5, its recall remains consistently low, indicating a clear problem of missed detections. As the IoU threshold increases, the $F1$-score drops sharply, further revealing its insufficient localization accuracy. Although YOLO models do not show an excessive tendency toward extreme strategies, their overall accuracy is still poor.

On the hard sample set, the performance of closed-set models degrades catastrophically: YOLO11's $F1$-score drops to 10.70% (IoU > 0.5), faster R-CNN's recall plummets to 5.11%, and RT-DETR even achieves zero detection performance on all hard sample categories. YOLOv8, as the benchmark for hard sample screening, maintains a balanced P and R (BI > 94%), but its absolute accuracy is extremely low ($F1$ = 6.01% at IoU > 0.5), which verifies that the four screened categories are indeed the most challenging defect types for traditional visual detectors.

It is worth noting that during the tuning process of these models under different confidence thresholds, they often sacrifice recall to obtain higher precision, leading to a large number of missed detections in actual industrial scenarios.

2) Open-Set Detection Models: High Recall but Unusable: As the most representative open-set detector currently, GroundingDINO shows strong semantic generalization ability in the open-set scenario. However, in this task, although the model achieves a recall rate as high as 93.97%, the precision is extremely low $P \approx 8\%$, resulting in an $F1$ value of less than 15%, and the BI is generally lower than 10%, reflecting an obvious tendency of "over-detection."

On the hard sample set, GroundingDINO models still maintain a relatively high recall, but their precision drops further, leading to a slight increase in $F1$-score but still below 15%. This extreme behavior is unacceptable in industrial quality inspection: although the model can capture more potential hard defects, a large number of FPs will significantly increase the cost of manual review and even mislead production decisions. Therefore, despite its advantage in recall on hard samples, GroundingDINO lacks control over detection quality and is difficult to deploy directly.

3) Our Method: Maintaining High Balance at Multiple Scales: In contrast, our method (based on Qwen3-VL-8B) shows excellent strategy balance and robustness on both the full sample set and the hard sample set under all three IoU thresholds. Specifically given as follows.

1) On the full sample set, the $F1$-score of our method is the highest among all methods at all IoU thresholds, and the BI is stably close to 100%, indicating that precision and recall are highly balanced and no extreme strategies are adopted.
2) On the hard sample set, our method still maintains perfect balance between P and R at all IoU thresholds, and its $F1$-score still remains at the optimal level. This fully verifies that our VTP-ICL mechanism with visually annotated example images and the two-stage reflection pipeline can effectively mitigate the performance degradation of the model on hard samples caused by difficulty bias.
3) The BPS metric, which jointly measures detection performance and strategy stability, shows that our method achieves the highest value under all IoU thresholds on both datasets, indicating that the model not only has good overall accuracy but also maintains a reasonable precision–recall tradeoff—this is a key advantage for industrial quality inspection scenarios with strict robustness requirements.

In any case, simply pursuing high precision or high recall to increase $F1$ cannot guarantee practical usability, especially on hard samples that are the main source of detection errors in industry. Our method achieves a breakthrough in this dimension: the visually annotated example images in

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TABLE II
QUANTITATIVE COMPARISON ON THE FULL SAMPLE SET UNDER FEW-SHOT SETTINGS

Type	Model	IoU>0.5					IoU>0.75					IoU>0.9				
		P	R	F1	BI	BPS	P	R	F1	BI	BPS	P	R	F1	BI	BPS
Closed-set	YOLO11m-pt	37.86	16.88	23.35	44.59	10.41	22.14	9.87	13.65	44.58	6.09	3.57	1.59	2.20	44.54	0.98
	YOLOv8m-pt	12.37	11.78	12.07	95.23	11.49	6.69	6.37	6.53	95.22	6.21	1.00	0.96	0.98	96.00	0.94
	Faster R-CNN	**90.57**	15.29	26.16	16.88	4.42	**64.15**	10.83	18.53	16.88	3.13	9.43	1.59	2.72	16.86	0.46
	RT-DETR	55.56	1.59	3.09	2.86	0.09	55.56	1.59	3.09	2.86	0.09	**33.33**	0.96	1.87	2.88	0.05
Open-set	GroundingDINO-Swin-T	10.53	93.55	18.93	11.26	2.13	8.64	76.77	15.53	11.25	1.75	5.09	37.33	8.96	13.64	1.22
	GroundingDINO-Swin-B	10.34	**97.99**	18.71	10.55	1.97	8.50	**88.89**	15.52	9.56	1.48	6.17	**54.84**	11.09	11.25	1.25
Ours	Qwen3-VL-8B	57.19	57.01	**57.10**	**99.69**	**56.92**	30.67	30.57	**30.62**	**99.67**	**30.52**	14.38	14.33	**14.35**	**99.65**	**14.31**

[a]Performance metrics (Precision, Recall, F1, Balance Index, and Balanced Performance Score) are reported in percentages (%) across three IoU thresholds (0.5, 0.75, 0.9). Best values are indicated in bold and second-best values are underlined. The proposed method (Qwen3-VL-8B) demonstrates superior balance and robustness compared to closed-set and open-set baselines.

TABLE III
QUANTITATIVE COMPARISON ON THE HARD SAMPLE SET UNDER FEW-SHOT SETTINGS

Type	Model	IoU>0.5					IoU>0.75					IoU>0.9				
		P	R	F1	BI	BPS	P	R	F1	BI	BPS	P	R	F1	BI	BPS
Closed-set	YOLO11m-pt	20.00	7.30	10.70	36.50	3.90	8.00	2.92	4.28	36.50	1.56	2.00	0.73	1.07	36.50	0.39
	YOLOv8m-pt	6.20	5.84	6.01	94.19	5.67	2.33	2.19	2.26	93.99	2.12	0.00	0.00	0.00	0.00	0.00
	Faster R-CNN	**70.00**	5.11	9.52	7.30	0.70	**30.00**	2.19	4.08	7.30	0.30	0.00	0.00	0.00	0.00	0.00
	RT-DETR	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Open-set	GroundingDINO-Swin-T	5.08	63.50	9.41	8.00	0.75	4.44	55.47	8.22	8.00	0.66	3.33	41.61	6.17	8.00	0.49
	GroundingDINO-Swin-B	8.02	**94.16**	14.78	8.52	1.26	6.46	**75.91**	11.91	8.51	1.01	4.35	**51.09**	8.02	8.51	0.68
Ours	Qwen3-VL-8B	48.91	48.91	**48.91**	**100**	**48.91**	29.93	29.93	**29.93**	**100**	**29.93**	**14.60**	14.60	**14.60**	**100**	**14.60**

[a]Results on the subset of defects characterized by low contrast and subtle textures, identified via the YOLOv8m-pt baseline (metrics reported in %). Best values are indicated in bold and second-best values are underlined. Note the catastrophic performance drop in traditional models versus the maintained stability of our approach.

VTP-ICL guide the model to focus on the real defect region at the visual perception stage, and the RM corrects the potential misdetections on subtle defects, enabling the model to achieve stable and reliable detection results in a single output on both easy and hard samples. This result fully verifies the effectiveness of the VTP-ICL and the two-stage reflection pipeline designed.

E. Ablation Study

To deeply understand the key design factors of the proposed method and their roles in the few-shot fabric defect detection task, we conduct the following three sets of ablation experiments.

1) Ablation of Method Components: We first explore the impact of each component in VTP-ICL and the RM. Specifically, we decompose the prompt strategy into four key components.

1) *Visual Demonstration (VD):* Natural language description generated by the model based on examples.
2) *Nonreasoning Output (noRO):* Directly generate structured results without additional reasoning processes.
3) *RM:* Verify and correct the initial output.
4) *Reasoning Output (RO):* Enable CoT, requiring the model to output coordinates after step-by-step reasoning.

The ablation settings and results are shown in Table IV: using Qwen3-VL-8B and Qwen3-VL-32B as the base, the performance under different combinations is tested, respectively. The results are as follows.

1) On the 8B model, when "Description + RO" is enabled, the $F1$ is only 19.90%; when changed to "Description + No RO," the $F1$ increases to 44.05%. This significant improvement verifies that directly generating structured output is superior to the explicit reasoning process.
2) More importantly, when the "Reflection" mechanism is further introduced (i.e., "Description + No RO + Reflection"), the $F1$ reaches 57.10%, an increase of more than 150% compared with the base version. This indicates that reflection can not only correct false detections but also enhance the model's robustness to complex texture backgrounds.

Key Finding 1: CoT contaminates structured parsing. Although large language models have strong language reasoning capabilities, in tasks involving precise numerical output, CoT introduces "estimative language" (e.g., "approximately at x = 200" and "close to y = 300"). These imprecise expressions replace the real coordinate values in the final output, leading to systematic deviations in the parsed bounding boxes. Since the model output is a sequence of discrete tokens, such ambiguous expressions cannot be accurately decoded into high-precision coordinates, thereby "contaminating" the token pool for structured parsing. Therefore, avoiding the use of CoT output is a prerequisite for achieving high-precision localization.

Key Finding 2: The wording of reflection prompts is crucial. We find that prompt design in the reflection stage decisively impacts performance. Prompts with strong modification instructions (e.g., "Please correct...") often cause the model to unnecessarily alter correct predictions, degrading performance. Conversely, neutral, verification-oriented wording (e.g., "Please review...") effectively suppresses excessive intervention, correcting only actual errors. Therefore, reflection prompts should emphasize "verification" rather than "correction" to maintain respect for high-quality

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TABLE IV
ABLATION STUDY ON PROMPT STRATEGIES AND RM (IOU > 0.5)

Model	Ablation Setting	P	R	F1
Qwen3-VL-8B	VD+RO(Base)	20.10	19.70	19.90
	noVD+noRO	44.48	43.63	44.05
	VD+noRO	45.10	43.95	44.52
	VD+noRO+RM	57.19	57.01	57.10
Qwen3-VL-32B	VD+RO(Base)	31.80	31.20	31.50
	noVD+noRO	40.96	38.22	39.54
	VD+noRO	45.05	42.04	43.94
	VD+noRO+RM	54.07	52.87	53.46

[a]Analysis of component contributions: Visual Demonstration (VD), Non-Reasoning Output (noRO), and Reflection Mechanism (RM). Results highlight the detriment of Reasoning Output(RO) on coordinate precision and the significant gain from the proposed reflection pipeline(metrics reported in %).

TABLE V
COMPARISON OF DIFFERENT REFLECTION PROMPT STYLES (IOU > 0.5)

Model	Reflection Prompt Style	P	R	F1
Qwen3-VL-8B	Verification-Oriented	57.19	57.01	57.10
	Modification-Oriented	50.64	50.32	50.48
Qwen3-VL-32B	Verification-Oriented	54.07	52.87	53.46
	Modification-Oriented	48.87	48.09	48.48

[a]Verification-oriented prompts consistently outperform modification-oriented ones across Qwen3-VL models(metrics reported in %).

TABLE VI
CROSS-ARCHITECTURE COMPARISON OF MLLMS (IOU > 0.5)

Model	P	R	F1	BI	BPS
Qwen3-VL-Plus	58.79	58.60	58.69	99.68	58.51
GLM-4.6V	53.33	50.96	52.12	95.56	49.80
Kimi-k2.5	44.55	42.99	43.76	96.50	42.22

[a]Performance of different foundation models under the proposed VTP-ICL and Reflection framework. Qwen3-VL-Plus achieves state-of-the-art balanced performance(metrics reported in %).

initial predictions. Table V illustrates the performance gap between these two distinct prompt styles, confirming that a verification-oriented approach consistently outperforms modification-oriented instructions.

2) Performance Comparison Among Different Models: We further compared the performance of three mainstream MLLMs under the same prompt strategy (shown in Table VI).

The results show that Qwen3-VL-Plus is significantly superior to other models in all metrics, especially leading obviously in $F1$ (58.69%) and BPS (58.51%). GLM-4.6v [43] performs the second best, while Kimi-k2.5 [44] is low in both precision and recall, which may be related to its data distribution or fine-tuning strategy in visual-language alignment training. This result indicates that not all large language models have the same few-shot fine-grained detection ability, and the cross-modal fusion mechanism and pretraining objectives inside the model have a profound impact on downstream tasks.

Although $F1$-scores vary slightly among the models, they all demonstrate stable detection behavior (BI > 95%) and maintain higher $F1$-scores than traditional models, without adopting extreme strategies that sacrifice one index to improve another. This indicates that current mainstream MLLMs already have good coordinated localization and classification capabilities under few-shot conditions. Through the ICL mechanism, they can achieve stable and balanced predictions without fine-tuning, reflecting the effectiveness and robustness of our method.

TABLE VII
IMPACT OF MODEL SIZE ON FEW-SHOT DETECTION (IOU > 0.5)

Model	P	R	F1	BI	BPS
Qwen3-VL-8B	57.19	57.01	57.10	99.69	56.92
Qwen3-VL-32B	54.07	52.87	53.46	97.78	52.28
Qwen3-VL-30B-A3B	55.70	54.46	55.07	97.77	53.85
Qwen3-VL-235B-A22B	63.58	63.38	63.48	99.69	63.28

[a]Comparison of Qwen3-VL series across parameter sizes. Interestingly, the 8B variant outperforms larger 32B/30B versions in fine-grained tasks, suggesting better prompt sensitivity and less overfitting to general domains(metrics reported in %).

3) Performance Changes of the Same Model Under Different Sizes: We further investigated the performance of the Qwen3-VL series models under different parameter scales (shown in Table VII). Surprisingly, the 8B model outperforms the 32B and 30B-A3B versions in both $F1$ and BPS.

Although the 235B-A22B model is far ahead in performance, the 8B model performs better than the larger 32B and 30B-A3B versions, which seems counterintuitive. We believe this phenomenon is mainly caused by the following two factors.

1) *Model Generalization Ability Versus Complexity:* It is crucial to note that our framework operates in a training-free manner, leveraging the frozen weights of the MLLMs without any task-specific fine-tuning. Under this strict setting, the interaction between the model and the downstream task is governed solely by the input context (prompt). We posit that the superior performance of the smaller model (8B) is a direct consequence of this constraint. For larger models (e.g., 32B/30B-A3B), the vast parameter space—which has not been explicitly optimized for the specific defect detection task—may result in a "dilution" of relevant features. The prompt (VTP-ICL) might only activate a sparse and suboptimal subset of neurons within the massive network, leading to inefficient knowledge utilization. Conversely, the smaller 8B model, with its more compact architecture, allows our structured visual–textual prompts to engage a larger effective proportion of its pretrained parameters. This higher degree of "prompt-to-parameter" alignment enables the 8B model to achieve better feature extraction in this few-shot downstream task.
2) *Differences in Prompt Sensitivity:* Large language models are usually more sensitive to prompt formats. If the prompt template fails to fully stimulate their potential, they may "fall silent" or generate lengthy and irrelevant content, thereby reducing parsing efficiency. In contrast, the 8B model responds more directly and outputs more concisely, making it more suitable for the current structured extraction task.

In addition, the excellent performance of 235B-A22B further proves that a larger model capacity can indeed help improve performance, but its cost is high, and deployment is difficult. Our method can achieve excellent results at the 8B scale, showing a good cost-performance advantage.

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V. Conclusion

This article presents a robust framework for few-shot fabric defect detection leveraging the semantic reasoning and contextual learning of frozen MLLMs. Addressing data scarcity and difficulty imbalance, our VTP-ICL and two-stage reflection pipeline bridge the gap between general foundation models and specialized industrial needs.

Experiments on a real-world dataset show our method significantly outperforms traditional closed-set and open-set baselines, achieving superior $F1$-scores and precision–recall balance. Crucially, ablation studies reveal that bypassing explicit CoT reasoning during coordinate generation, in favor of a dedicated reflection stage, is vital for numerical accuracy. Repositioning reasoning from intermediate generation to postprediction verification neutralizes noise from step-by-step thinking, transforming it into a self-correction asset. Furthermore, smaller models (e.g., 8B parameters) outperforming larger ones highlights the importance of prompt sensitivity and task alignment over parameter scale. While demonstrating MLLMs' viability in low-resource industrial scenarios, future work will focus on optimizing inference latency for real-time deployment and exploring automated prompt generation. This study paves the way for generalizable, interpretable, and data-efficient intelligent inspection systems adaptable to modern manufacturing.

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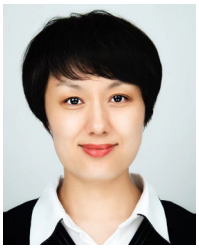
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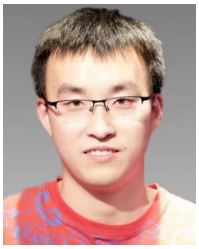
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